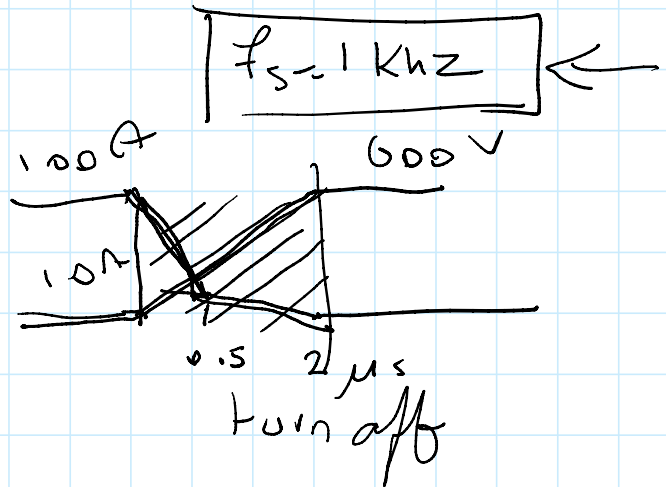
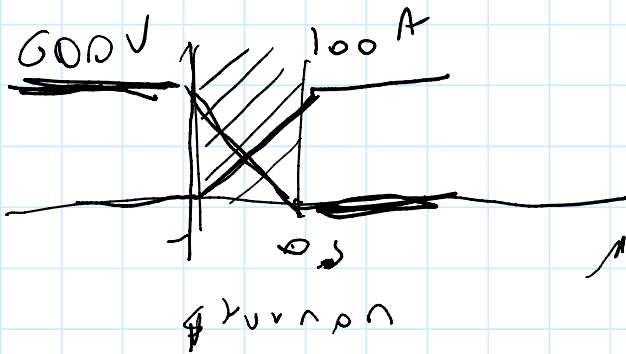


05-03-12

Tuesday, June 2, 2026 8:39 AM

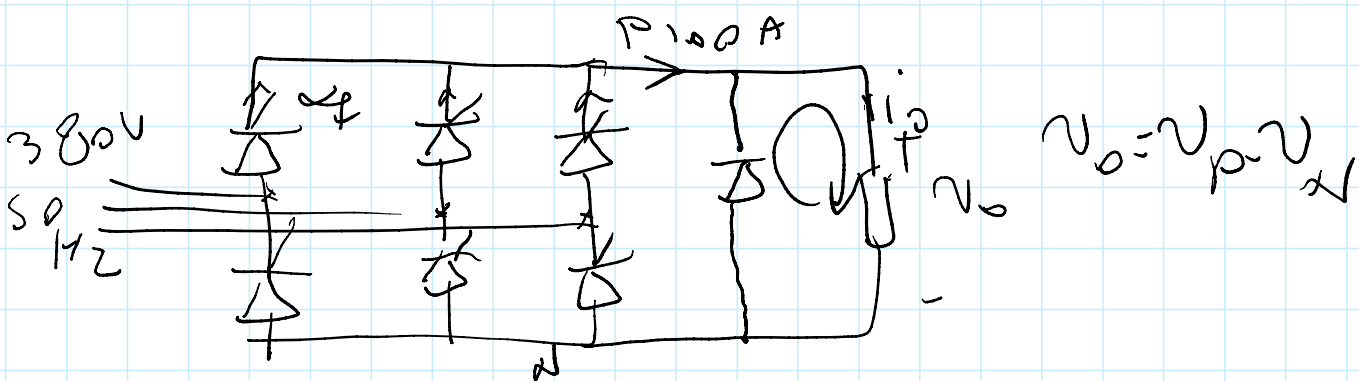


a factory needs a controllable dc-supply
- no inverter operation

load i_{ac} i_{dc} $0 \leq \alpha < 90$ $\bar{V}_o > 0$
 > 90 $\bar{V}_o < 0$
input: 3 ϕ 380V 50Hz

Γ	$n\bar{V}$	$\pm \checkmark$	Rect.
Π	Π	$\checkmark$	Inver.

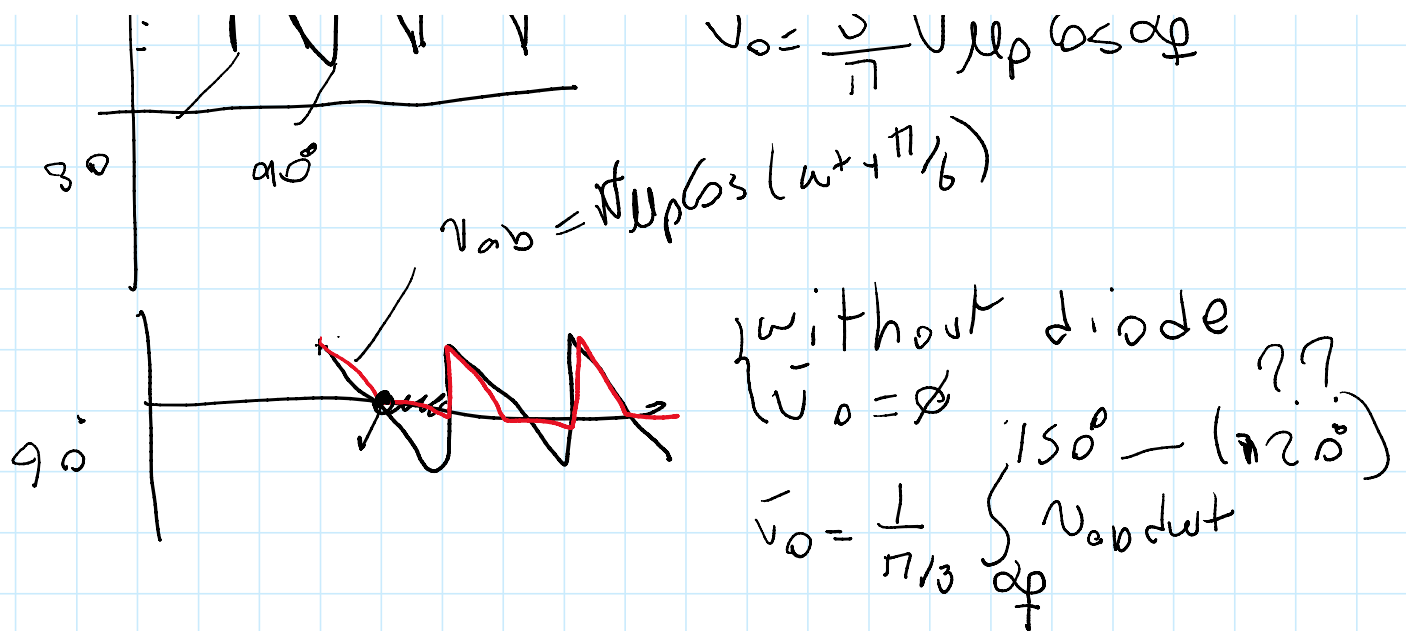
a) draw the power circuit of the proposed converter.



b) $\bar{I}_o = 100 \text{ A}$, $\bar{V}_o$ $\left\{ \begin{array}{l} \phi = 30^\circ \quad \bar{V}_o = ? \\ \phi = 90^\circ \quad \bar{V}_o = ? \end{array} \right.$



$$\bar{V}_o = \frac{3}{\pi} V_{mp} \cos \phi$$



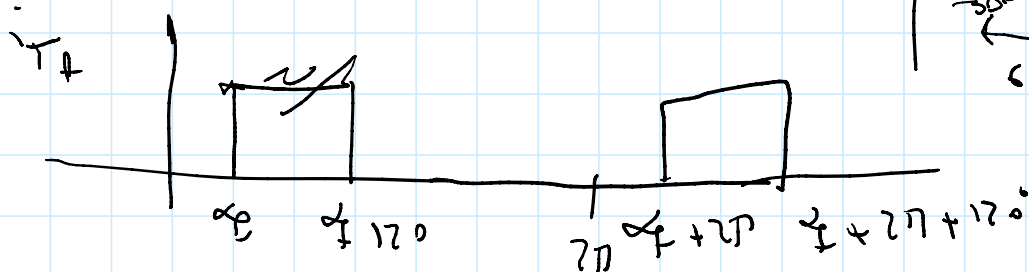
c) select the switches:

$V_{rem} = V_{orm} \leq 5 P. V_{mp} = 1.5 \times 380\sqrt{2}$

$\bar{I}_{Tav} = \frac{I_{oma}}{3} \times S.f. = \frac{100}{3} \approx 1.5$

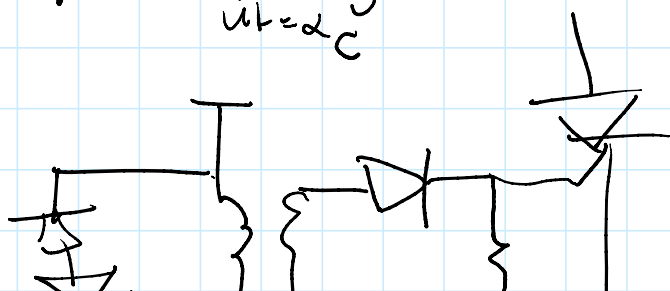
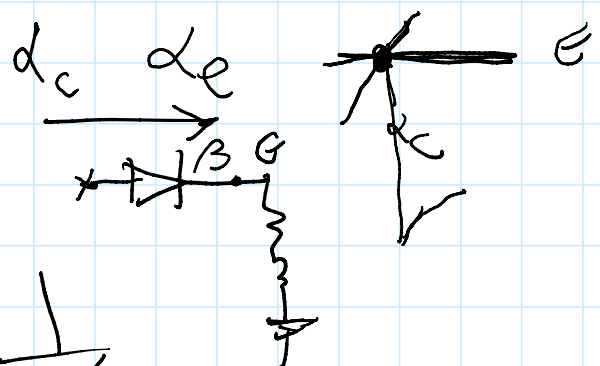
$\bar{V}_o = \frac{3}{\pi} V_{mp} \cos \alpha$

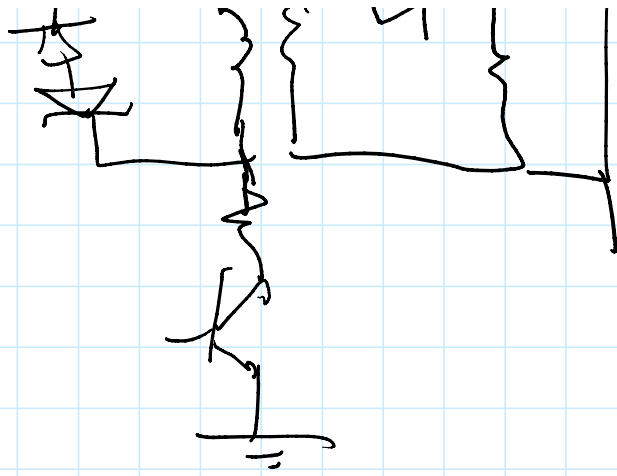
Type: phase-controlled.



$K_c \leq \frac{\sqrt{3}}{2} \text{ (ridge)}$

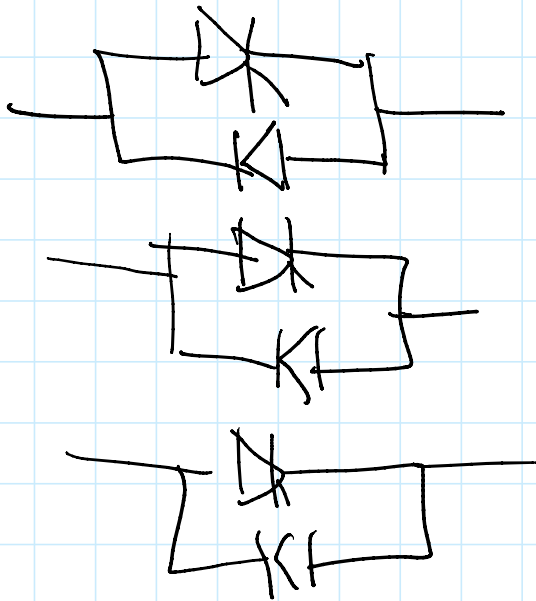
$V_{mp} \sin(\omega t + \pi/6) = E$
 $\omega t = \alpha_c$



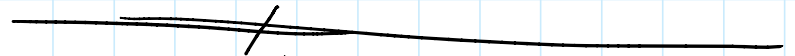


دکترایانند بر سر راه است؟

میت 2, IV



سوال ✓
حالم : ✓
زبان



به صورت کلی
در PC و DSP
به صورت دیجیتال و آنالوگ

اینها مدیا

دوین
میدان
دوین
dual
motor control